Division: Finding the Unknown Factor
Answer Key
Grade 3

Quick Answers

1. 4
2. 6
3. 6 stickers
4. 6
5. 7
6. 5 boxes
7. 6
8. 8
9. (b) = 5 rows, —
10. apples in each bag = 9 apples, —

What is verified

8 of 10 problems fully machine-checked against SymPy.
— marks an answer only you can judge — problems 9, 10. The worked solution states what a correct response must contain.

Curriculum

Level: Grade 3
3.OA.A.2 – *problems 2, 5, 8*
3.OA.A.3 – *problems 3, 6, 9, 10*
3.OA.A.4 – *problems 1, 4, 7, 9*
Difficulty 1–4 of 5 across 12 tagged checks.

Common wrong answers

6. *If they got 24: subtracted instead of dividing*

1. Find the missing factor: $5 \times n = 20$
Solution: Ask: 5 times what makes 20? From the fives table, $5 \times 4 = 20$. $\boxed{n = 4}$

2. $18 \div 3 = ?$ Think: $3 \times \mathbf{?} = 18$.
Solution: Division asks for the unknown factor: 3 times what makes 18? Since $3 \times 6 = 18$, we get $18 \div 3 = 6$. $\boxed{6}$

3. 24 stickers placed equally onto 4 pages. How many on each page?

Solution: Sharing 24 into 4 equal groups asks $4 \times \textbf{?} = 24$. The related fact is $4 \times 6 = 24$, so each page gets 6 stickers. $\boxed{\text{6 stickers}}$

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4. Find the missing factor: $n \times 7 = 42$

Solution: What number of 7s makes 42? From the sevens table, $6 \times 7 = 42$. $\boxed{n = 6}$

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5. $35 \div 5 = ?$ Think: $5 \times \textbf{?} = 35$.

Solution: 5 times what makes 35? $5 \times 7 = 35$, so $35 \div 5 = 7$. $\boxed{7}$

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6. 30 markers, 6 in each box. How many boxes?

Solution: Making groups of 6 from 30 asks $6 \times \textbf{?} = 30$. Since $6 \times 5 = 30$, she fills 5 boxes. (Subtracting, $30 - 6 = 24$, only removes one box's markers — it does not count boxes.) $\boxed{\text{5 boxes}}$

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7. Find the missing factor: $9 \times n = 54$

Solution: 9 times what makes 54? From the nines table, $9 \times 6 = 54$. $\boxed{n = 6}$

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8. $48 \div 6 = ?$ Think: $6 \times \textbf{?} = 48$.

Solution: 6 times what makes 48? $6 \times 8 = 48$, so $48 \div 6 = 8$. $\boxed{8}$

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9. 40 chairs, 8 chairs in each row.

Solution (a): The story fixes the group size (8) and hides the number of rows, so the fact is $8 \times \textbf{?} = 40$ (also accepted: $\textbf{?} \times 8 = 40$ or $8 \times n = 40$).

Solution (b): 8 times what makes 40? $8 \times 5 = 40$, so she makes 5 rows. $\boxed{\text{5 rows}}$

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10. Challenge: 63 apples in 7 equal bags. How does $7 \times 9 = 63$ help?

Solution: Splitting 63 apples equally into 7 bags asks the missing-factor question "7 times what makes 63?" The known fact $7 \times 9 = 63$ answers it directly: the missing factor is 9, so $63 \div 7 = 9$ and each bag holds 9 apples. A correct explanation must connect the division to the known multiplication fact, not just state the answer. $\boxed{\text{9 apples}}$